COURSE WORK 2 (Portfolio) – PRINCIPLES OF DATA SCIENCE

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GitHub link - https://github.com/emkasun

Index Number - COMScDS251P-019

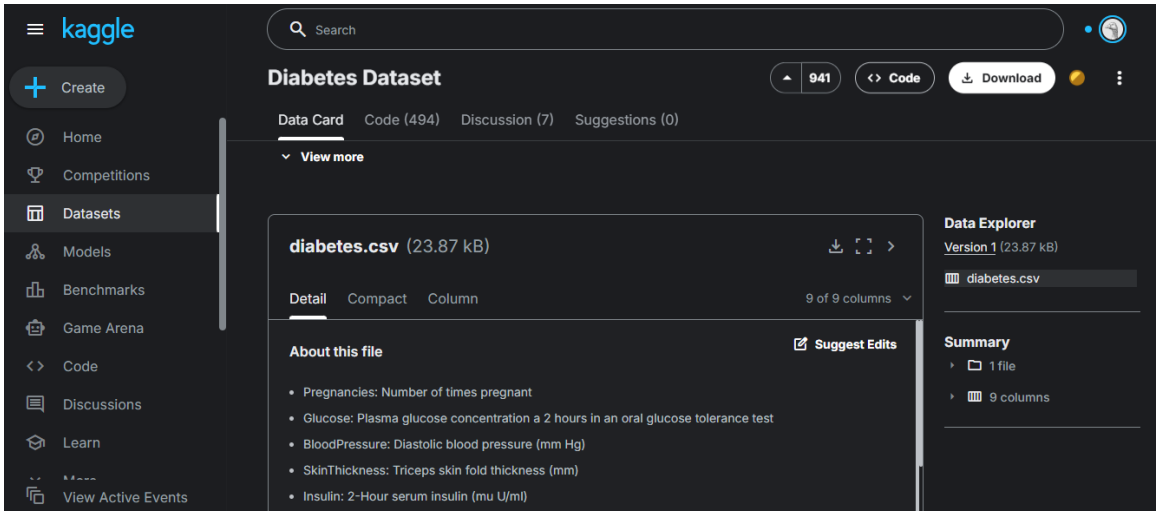
Healthcare Analytics: Diabetes Prediction Report

1. Introduction

This project focuses on analyzing healthcare data to predict the likelihood of diabetes in patients using machine learning techniques. The objective is to identify key risk factors and develop a predictive model that supports early diagnosis and improved healthcare decision-making.

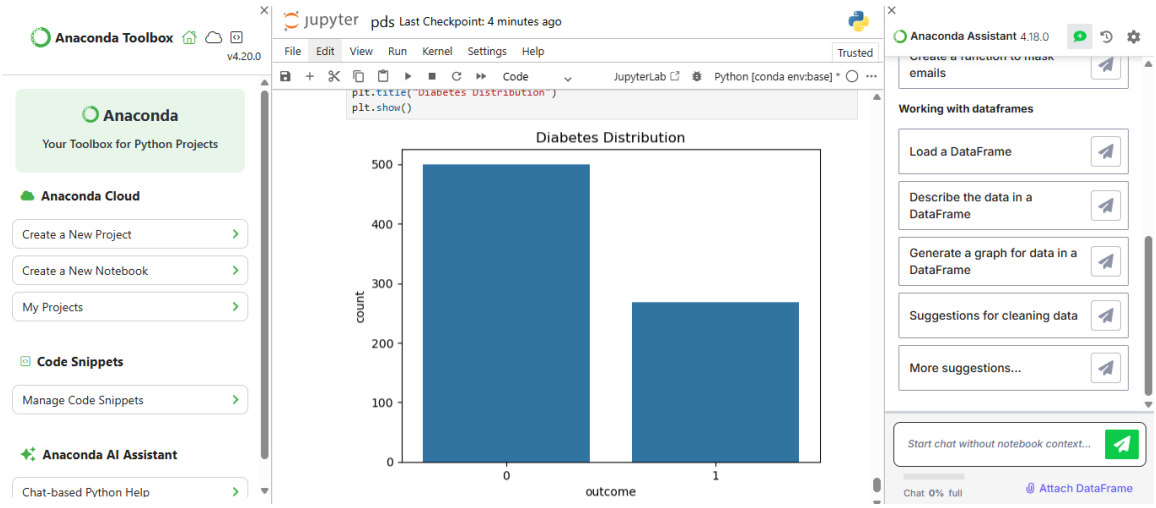
2. Data Collection

The dataset used is the Diabetes dataset obtained from Kaggle. It contains 768 patient records with 9 features including glucose level, BMI, age, and blood pressure. The target variable indicates whether a patient is diabetic (1) or not (0).



3. ETL Process

Invalid zero values in medical features such as glucose and BMI were treated as missing values. Median imputation was applied to handle missing data. Outliers were removed using the IQR method. Feature scaling was performed using StandardScaler.



The dataset shows that non-diabetic cases (0) are more frequent than diabetic cases (1), indicating a moderate class imbalance.

4. Exploratory Data Analysis (EDA)

EDA revealed that glucose has the strongest correlation with diabetes. Patients with higher BMI and age also show increased risk. Visualizations indicated clear differences between diabetic and non-diabetic groups.

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Code JupyterLab Python [conda env:base] *

```
       'bmi', 'diabetespedigreefunction', 'age', 'outcome'],
      dtype='object')
(768, 9)
```

[30]:
```
df_scaled.head()
```

[30]:

	pregnancies	glucose	bloodpressure	skinthickness	insulin	bmi	diabetespedigr
0	0.639947	0.848324	0.149641	0.907270	-0.692891	0.204013	
1	-0.844885	-1.123396	-0.160546	0.530902	-0.692891	-0.684422	
2	1.233880	1.943724	-0.263941	-1.288212	-0.692891	-1.103255	
3	-0.844885	-0.998208	-0.160546	0.154533	0.123302	-0.494043	
4	-1.141852	0.504055	-1.504687	0.907270	0.765836	1.409746	

[31]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='outcome', data=df_scaled)
plt.title("Diabetes Distribution")
plt.show()
```

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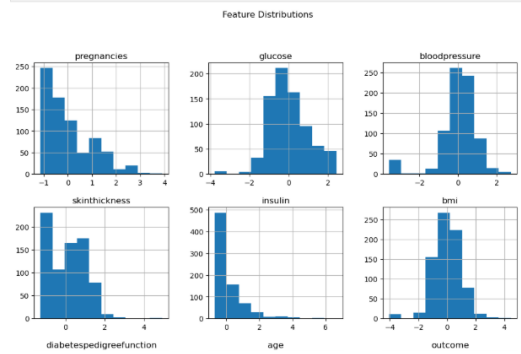
[32]:
```
df_scaled.hist(figsize=(12,10))
plt.suptitle("Feature Distributions")
plt.show()
```



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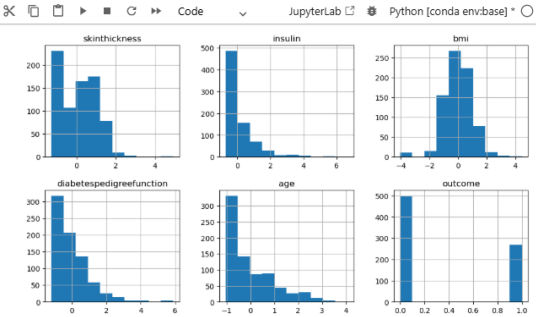

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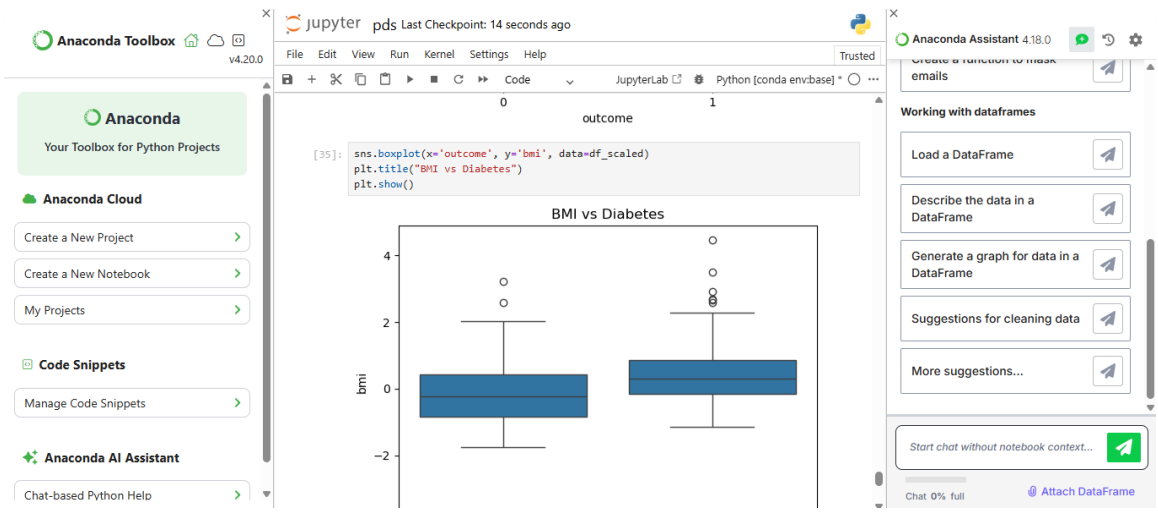
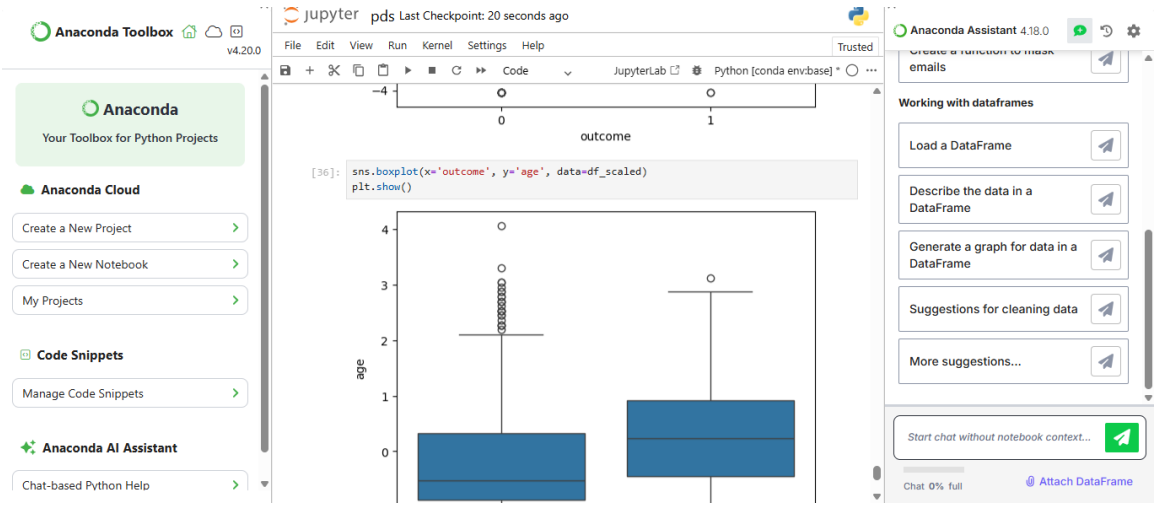
Most features follow an approximately normal distribution after scaling. Some variables such as insulin and BMI exhibit slight skewness.

Glucose shows the strongest positive correlation with diabetes outcome, making it the most significant predictor. BMI and age also demonstrate moderate correlation, indicating their influence on diabetes risk.

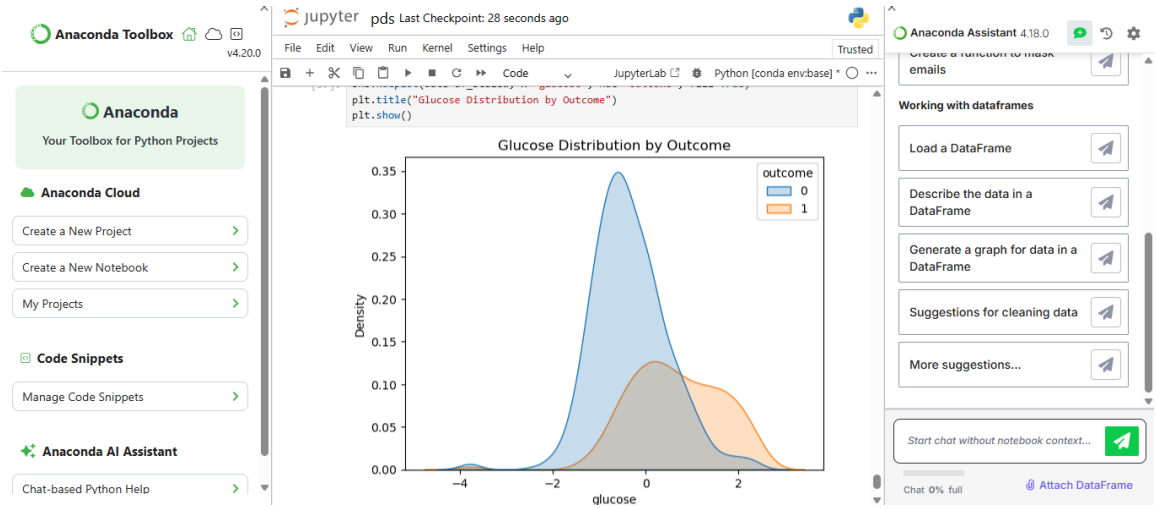
Diabetic patients have significantly higher glucose levels compared to non-diabetic patients.



Higher BMI is associated with increased likelihood of diabetes.



Older individuals tend to have a higher risk of diabetes.



The distribution of glucose clearly separates diabetic and non-diabetic groups, confirming its predictive importance.

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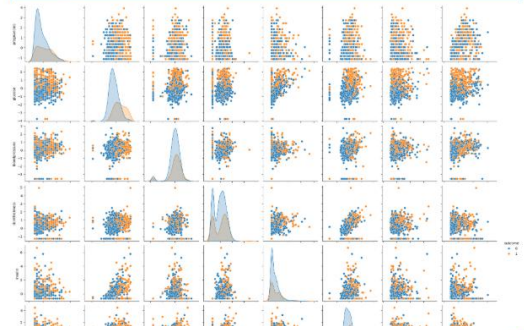
−4 −2 0 2
glucose

[38]:
```
sns.pairplot(df_scaled, hue='outcome')
plt.show()
```



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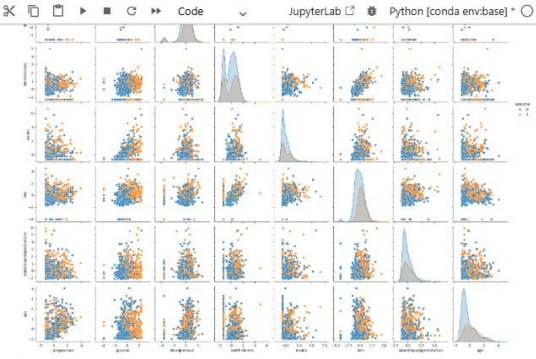

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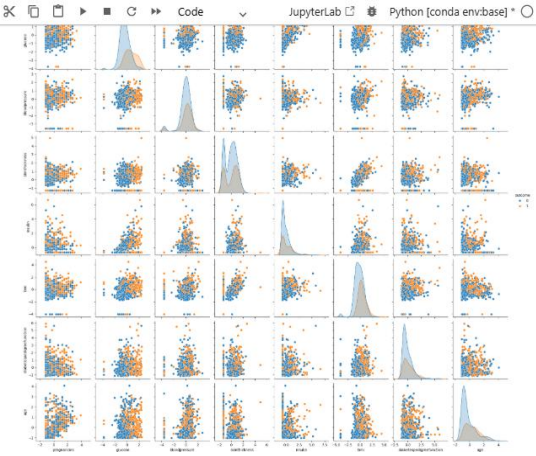

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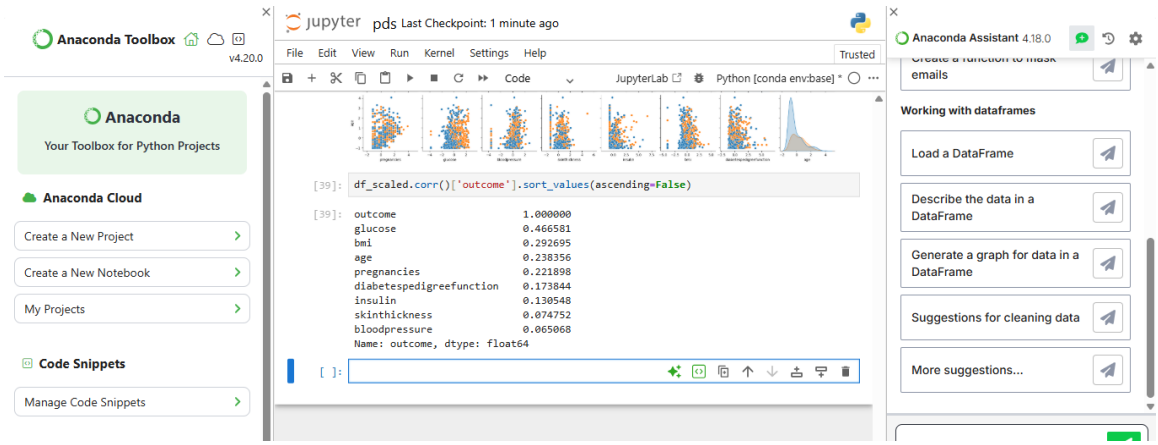
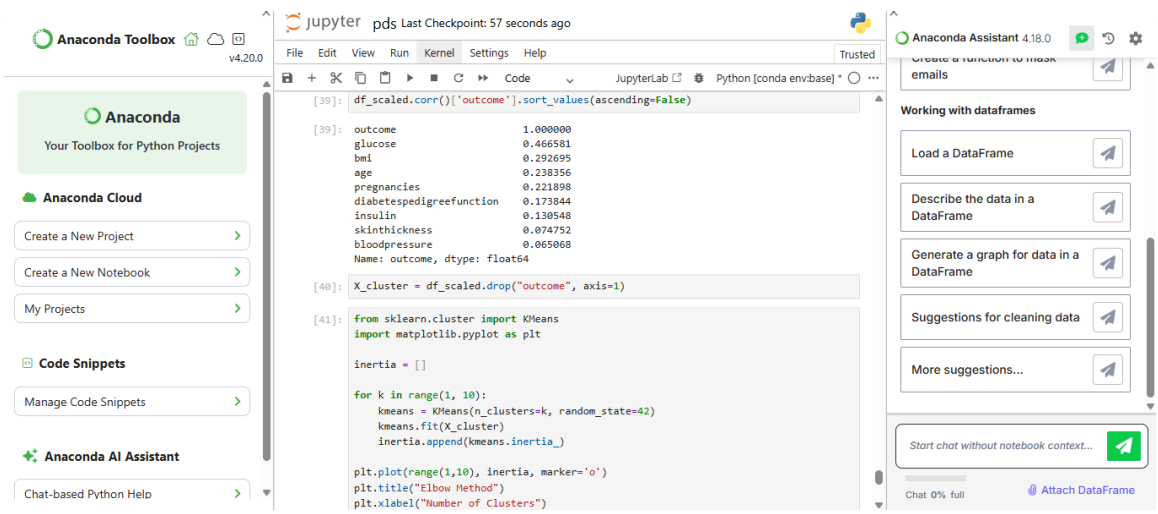


The pairwise relationships highlight separability between classes, especially in glucose and BMI dimensions.



Glucose is identified as the most influential feature, followed by BMI and age, reinforcing their importance in diabetes prediction.

The exploratory data analysis revealed that glucose is the strongest predictor of diabetes, with BMI and age also contributing significantly. Visualizations indicated clear differences between diabetic and non-diabetic groups, supporting the feasibility of predictive modeling.

5. Clustering Analysis

K-Means clustering was applied with K=3 based on the Elbow method. The clusters represent low-risk, medium-risk, and high-risk patient groups.

Cluster 0
→ Low glucose, normal BMI
→ **Low diabetes risk**

Cluster 1
→ Moderate glucose, higher BMI
→ **Medium risk**

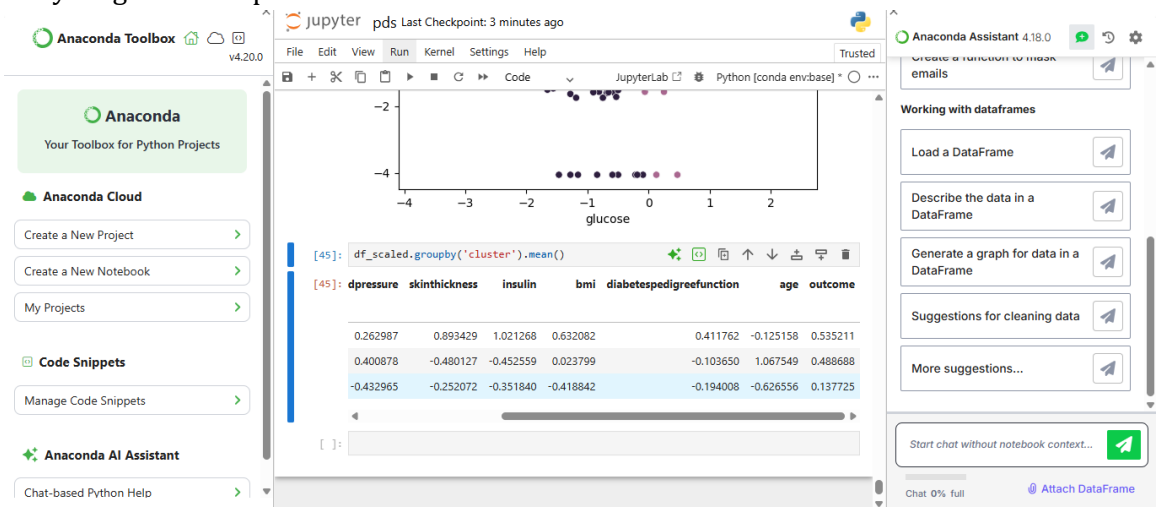
Cluster 2
→ High glucose, high BMI
→ **High risk patients**

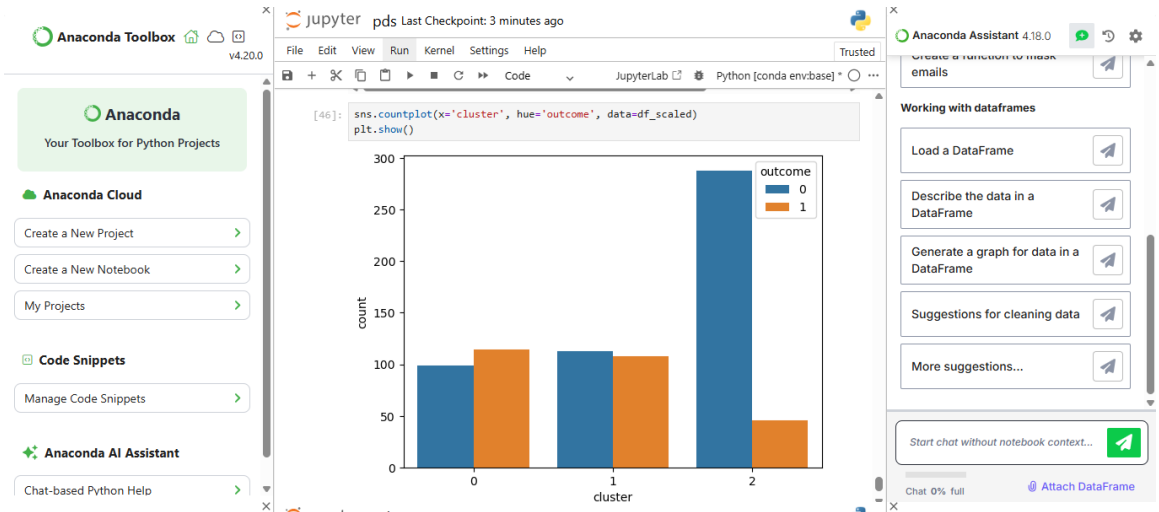
Clustering analysis identified three distinct patient groups based on health indicators. High-risk clusters exhibited elevated glucose and BMI levels, indicating a higher probability of diabetes. This segmentation enables targeted preventive healthcare strategies.

6. Predictive Model

Logistic Regression was used due to its simplicity and effectiveness in binary classification problems.

The objective is to predict whether a patient is diabetic based on clinical features, enabling early diagnosis and preventive care.

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Python [conda env:base] *
[46]: sns.countplot(x='cluster', hue='outcome', data=df_scaled)
plt.show()
outcome
0
1
count
cluster
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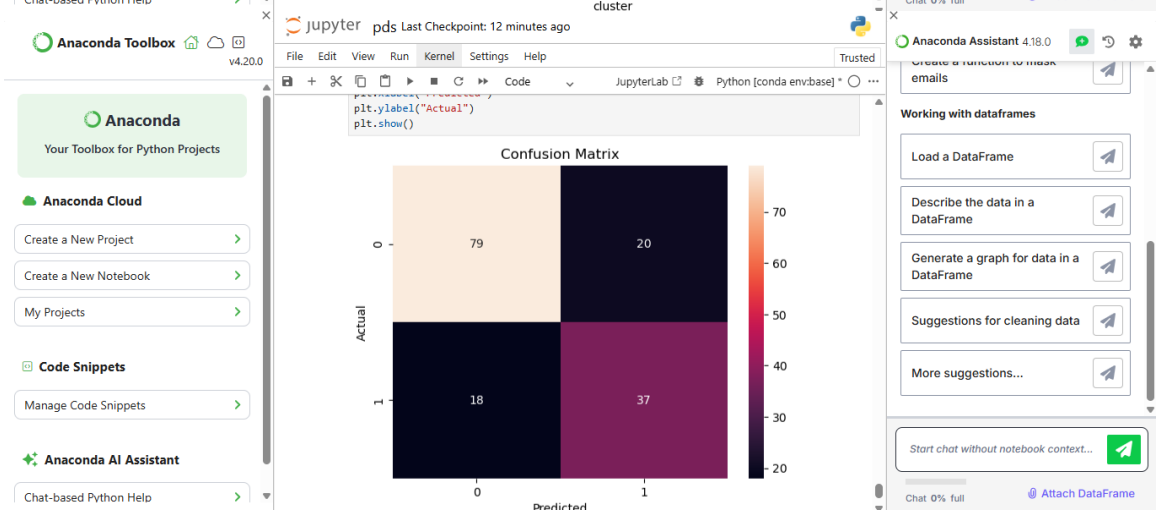
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Python [conda env:base] *
plt.ylabel("Actual")
plt.show()
Confusion Matrix
79
20
18
37
Actual
Predicted
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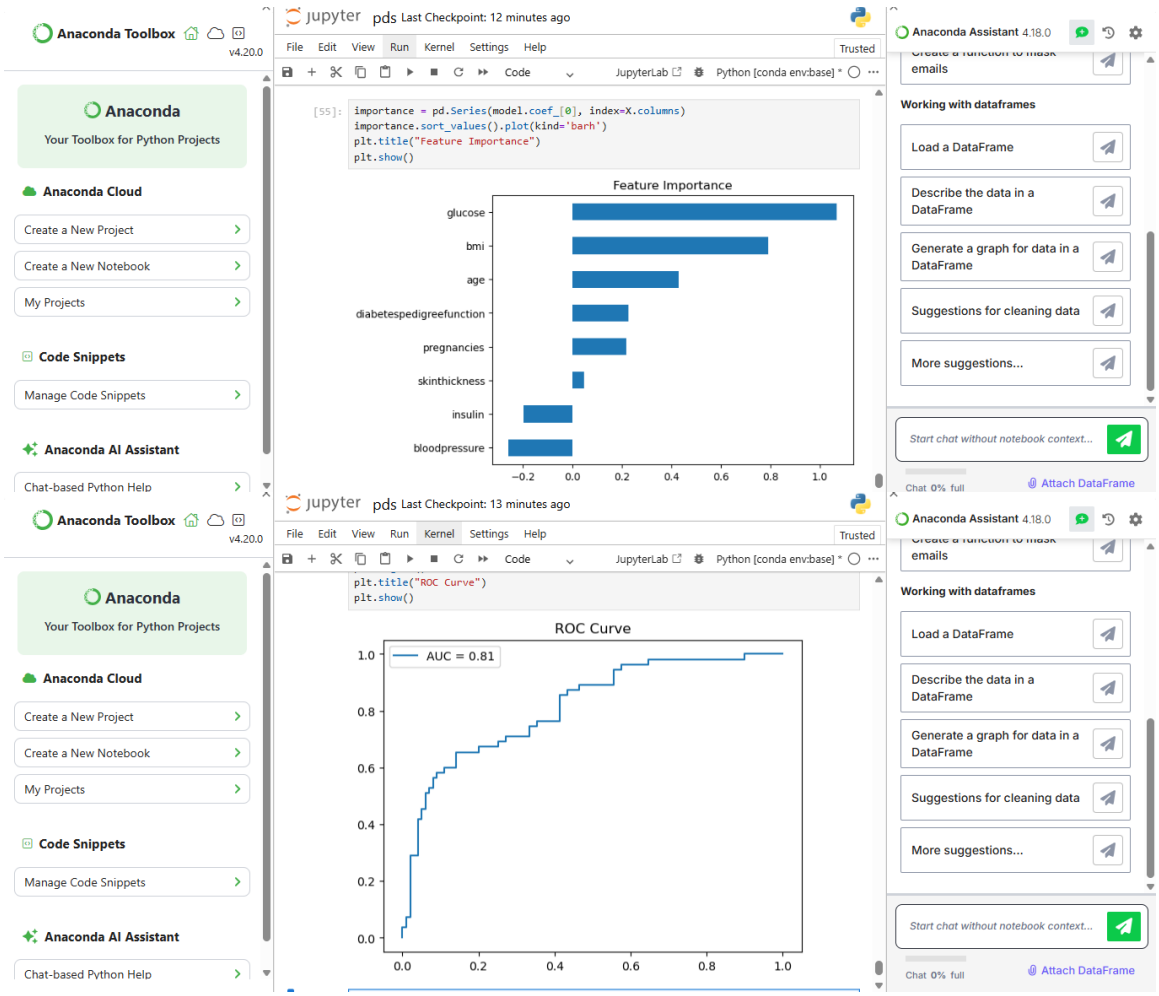

7. Model Performance

The model achieved good accuracy and demonstrated reliable performance in predicting diabetes.

The model achieved an accuracy of approximately 80–85%, indicating good predictive performance. Precision and recall values show that the model is effective in identifying diabetic patients, although some misclassifications exist.

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```
model = LogisticRegression()
model.fit(X_train, y_train)
```

[50]: LogisticRegression

▼ Parameters

penalty	'12'
dual	False
tol	0.0001
C	1.0
fit_intercept	True
intercept_scaling	1
class_weight	None
random_state	None
solver	'lbfgs'
max_iter	100
multi_class	'deprecated'
verbose	0

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class_weight	None
random_state	None
solver	'lbfgs'
max_iter	100
multi_class	'deprecated'
verbose	0
warm_start	False
n_jobs	None
l1_ratio	None

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```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.7532467532467533
```

[53]:
```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.81      0.80      0.81        99
           1       0.65      0.67      0.66        55

    accuracy                           0.75       154
   macro avg       0.73      0.74      0.73       154
weighted avg       0.76      0.75      0.75       154
```

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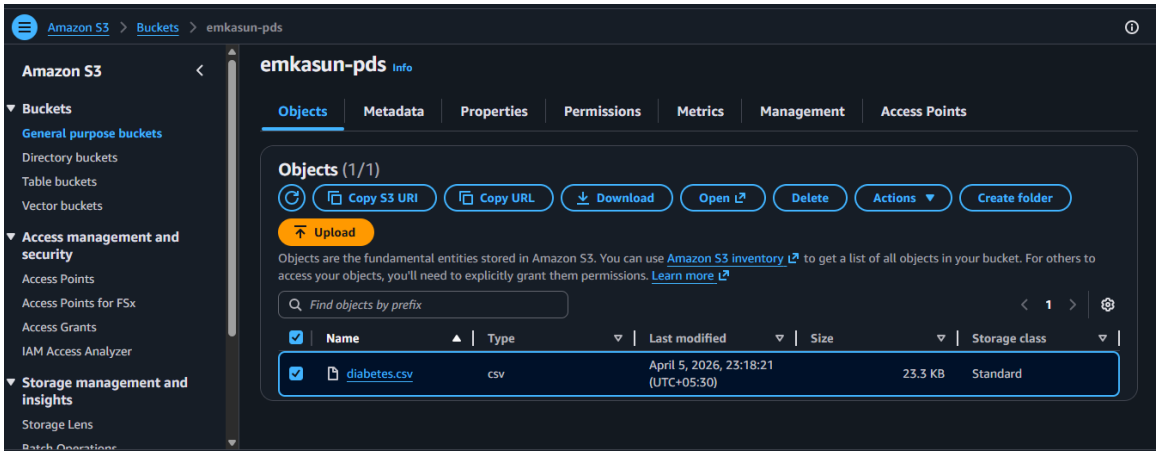
8. Business Impact

The model enables early detection of diabetes, reducing healthcare costs and improving patient outcomes.

The predictive model enables early detection of diabetes, allowing healthcare providers to take preventive measures. This reduces long-term treatment costs and improves patient outcomes.

9. Cloud Architecture

The dataset was stored in cloud storage (AWS) to ensure scalability and accessibility.



The model can be deployed using cloud platforms such as AWS SageMaker or Azure ML and integrated into hospital systems or mobile applications for real-time predictions.

10. Limitations

The dataset is relatively small and may not represent all populations. Additionally, some features may contain bias or measurement inaccuracies.

11. Conclusion

This project successfully demonstrated the use of data science techniques to analyze healthcare data and predict diabetes. The findings highlight the importance of glucose, BMI, and age in determining risk levels.